

Repeated Measures & Within-Designs

Statistical Methods Dossier | Standard Level

1. Method Description

The repeated measures design measures same subjects across time/conditions. Controls for inter-individual differences, reducing error variance. Three main approaches:

rm-ANOVA: Sphericity assumption (Mauchly test). Greenhouse-Geisser/Huynh-Feldt correction if violated.

MANOVA for RM: No sphericity assumption, less power for small n.

Mixed Models: Most flexible. Random intercepts per subject. Handles unbalanced data, missing values, arbitrary covariance structures.

2. Field Examples

Medicine: Blood pressure at days 0,7,14,28. Mixed model with AR1 covariance.

Psychology: Cognitive testing before/after sleep deprivation. rm-ANOVA, sphericity violated -> GG correction.

Sports: Maximal strength across 8 weeks training. Mixed model with random slopes.

Economics: Job satisfaction before/3mo/12mo after reorganization. Mixed model handles missing data.

3. Assumptions

rm-ANOVA: sphericity, normal differences, independence between subjects

Mixed Model: normal residuals, linearity, independent between-subject errors

4. Interpretation and Reporting

rm-ANOVA: F, df, p, partial η^2 , sphericity correction

Mixed Model: random effects variance (ICC), fixed effects (beta, SE, p), covariance structure

5. Code Examples

R:

```
lmer(wert ~ zeit*gruppe + (1|id), data=df)
lme(wert~zeit+gruppe, random=~zeit|id, correlation=corAR1(), data=df)
```

Python:

```
mixedlm("wert ~ zeit*gruppe", data=df, groups=df["id"])
```

6. Extension

Covariance structures: CS, AR1, unstructured. Compare via AIC/BIC.

Power analysis: G*Power for repeated measures ANOVA.

7. Exercises

1. Reaction time before/1h/4h after caffeine (n=30). Design? Method? Covariance structure?
2. Mauchly $p=.03$, 4 time points. Interpretation? Correction? df change?
3. ICC=0.45 in mixed model. Meaning? Advantage over rm-ANOVA with 20% missing?